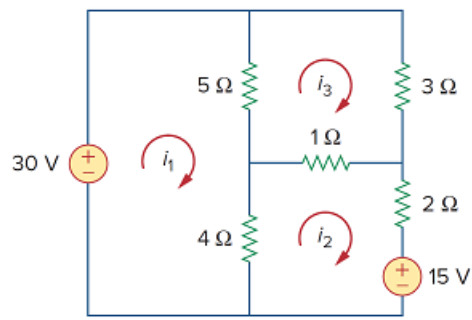


Problem

Write the mesh-current equations for the circuit in Fig. 3.115. Next, determine the values of i_1 , i_2 , and i_3 .

Figure 3.115:



Step-by-step solution

Step 1 of 8

Consider the following circuit diagram:

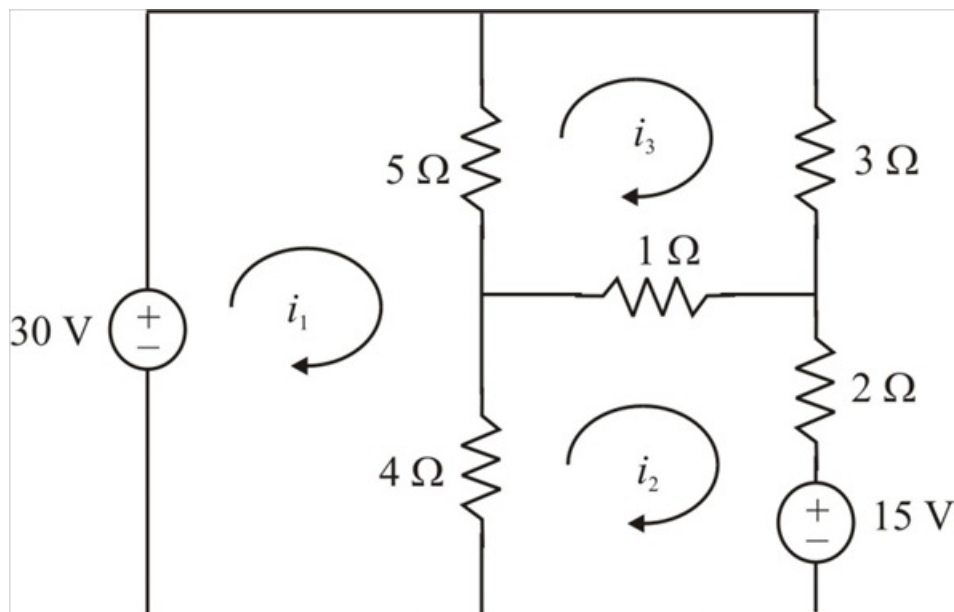


Figure 1

Step 2 of 8

If a circuit has N meshes, the mesh-current equations can be expressed in terms of resistances:

$$\begin{pmatrix} R_{11} & \cdots & R_{1N} \\ \vdots & \ddots & \vdots \\ R_{N1} & \cdots & R_{NN} \end{pmatrix} \begin{pmatrix} i_1 \\ \vdots \\ i_N \end{pmatrix} = \begin{pmatrix} v_1 \\ \vdots \\ v_N \end{pmatrix} \text{ or simply } \mathbf{R}\mathbf{i} = \mathbf{v}.$$

Where,

Sum of the resistances in mesh k is R_{kk} ,

Negative of the sum of the resistances in common with meshes k and j , $k \neq j$ is $R_{kj} = R_{jk}$

Unknown mesh current for mesh k in the clockwise direction is i_k

Sum taken clockwise of all independent voltage sources in mesh k , with voltage rise treated as positive is v_k .

Step 3 of 8

The circuit in Figure 1 has five meshes, so we need five mesh equations. This implies that the size of the resistance matrix $\mathbf{R}$, is 5 by 5.

Each of the diagonal terms is the sum of the resistances in the related mesh, while each of the off-diagonal terms is the negative of the resistances common to meshes.

Calculate the diagonal terms of $\mathbf{R}$, in ohms:

$$\begin{aligned} R_{11} &= 5 + 4 \\ &= 9 \end{aligned}$$

$$\begin{aligned} R_{22} &= 1 + 2 + 4 \\ &= 7 \end{aligned}$$

$$\begin{aligned} R_{33} &= 3 + 1 + 5 \\ &= 9 \end{aligned}$$

Find the off-diagonal terms in the first row:

$$R_{12} = -4$$

$$R_{13} = -5$$

Find the off-diagonal terms in the second row:

$$R_{21} = -4$$

$$R_{23} = -1$$

Find the off-diagonal terms in the third row:

$$R_{31} = -5$$

$$R_{32} = -1$$

Step 4 of 8

The input voltage vector $\mathbf{v}$ has the following terms, in volts:

$$v_1 = 30$$

$$v_2 = -15$$

$$v_3 = 0$$

Therefore, the mesh-current equations are

$$\begin{pmatrix} 9 & -4 & -5 \\ -4 & 7 & -1 \\ -5 & -1 & 9 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \\ i_3 \end{pmatrix} = \begin{pmatrix} 30 \\ -15 \\ 0 \end{pmatrix}.$$

Step 5 of 8

Use Cramer's rule to find determinant Δ of the matrix:

$$\begin{aligned}\Delta &= \begin{vmatrix} 9 & -4 & -5 \\ -4 & 7 & -1 \\ -5 & -1 & 9 \end{vmatrix} \\ &= 9(63-1) + 4(-36-5) - 5(4+35) \\ &= 558 - 164 - 195 \\ &= 199\end{aligned}$$

Use Cramer's rule to find determinant Δ_1 of the matrix:

$$\begin{aligned}\Delta_1 &= \begin{vmatrix} 30 & -4 & -5 \\ -15 & 7 & -1 \\ 0 & -1 & 9 \end{vmatrix} \\ &= 30(63-1) + 15(-36-5) + 0 \\ &= 1860 - 615 \\ &= 1245\end{aligned}$$

Use Cramer's rule to find determinant Δ_2 of the matrix:

$$\begin{aligned}\Delta_2 &= \begin{vmatrix} 9 & 30 & -5 \\ -4 & -15 & -1 \\ -5 & 0 & 9 \end{vmatrix} \\ &= 9(-135+0) + 4(270+0) - 5(-30-75) \\ &= -1215 + 1080 + 525 \\ &= 390\end{aligned}$$

Use Cramer's rule to find determinant Δ_3 of the matrix:

$$\begin{aligned}\Delta_3 &= \begin{vmatrix} 9 & -4 & 30 \\ -4 & 7 & -15 \\ -5 & -1 & 0 \end{vmatrix} \\ &= 9(0-15) + 4(0+30) - 5(60-210) \\ &= -135 + 120 + 750 \\ &= 735\end{aligned}$$

Step 6 of 8

Write the mesh current as follows:

Calculate the mesh current i_1 :

$$\begin{aligned}i_1 &= \frac{\Delta_1}{\Delta} \\ &= \frac{1245}{199} \\ &= 6.25 \text{ A}\end{aligned}$$

Therefore, the current i_1 is 6.25 A.

Step 7 of 8

Calculate the mesh current i_2 :

$$\begin{aligned}i_2 &= \frac{\Delta_2}{\Delta} \\ &= \frac{390}{199} \\ &= 1.959 \text{ A}\end{aligned}$$

Therefore, the current i_2 is 1.959 A.

Calculate the mesh current i_3 :

$$\begin{aligned} i_3 &= \frac{\Delta_3}{\Delta} \\ &= \frac{735}{199} \\ &= 3.694 \text{ A} \end{aligned}$$

Therefore, the current i_3 is 3.694 A .